

CPE 381 CLASSWORK 02: MATHEMATICAL PRELIMINARIES

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes

Due: August 26, 2026 (In-Class)

1. Consider the complex number z on Figure 1. What is the complex conjugate $\bar{z}$? (3 Points)

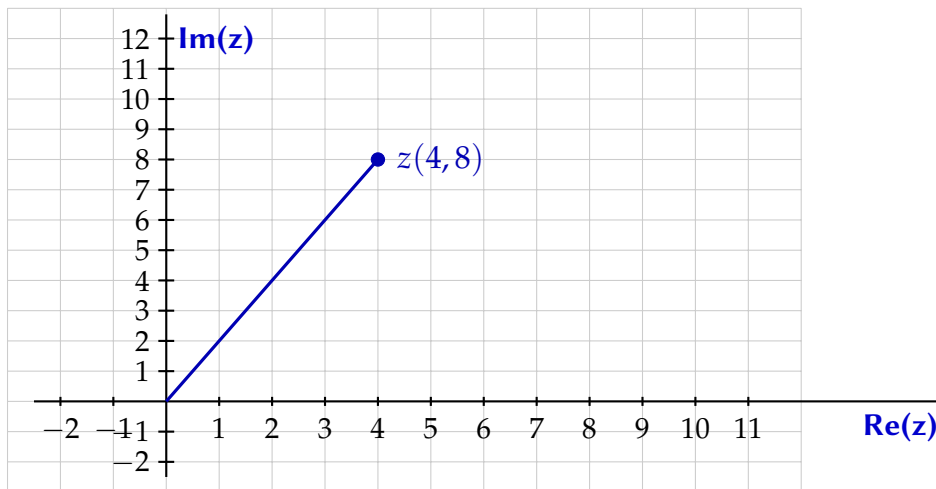


Figure 1: Complex z .

Answer

Complex conjugate of a complex number $z = a + jb$ is $\bar{z} = a - jb$. Hence for the given complex number, the complex conjugate is $\bar{z} = 4 - j8$.

2. If $z = \sqrt{3} + i$, what is z^4 in polar form? Use De Moivre's Theorem. (6 Points)

Answer

According to De Moivre's Theorem, if a complex number z is in its polar form $z = r(\cos \theta + j \sin \theta)$, then $z^n = r^n(\cos(n\theta) + j \sin(n\theta))$.

For the given z , $r = \sqrt{(\sqrt{3})^2 + 1^2} = \sqrt{4} = 2$, and $\theta = \tan^{-1} \frac{1}{\sqrt{3}} = \frac{\pi}{6}$ because z lies in the first quadrant. **(Note: if we had $z = \sqrt{3} - i$, then θ would not be $\frac{\pi}{6}$).**

Thus, $z^4 = 2^4(\cos(\frac{4\pi}{6}) + j \sin(\frac{4\pi}{6})) = 16(-\frac{1}{2} + j\frac{\sqrt{3}}{2}) = -8 + j8\sqrt{3}$.

3. What is the numerical value of $2(\sin^2 \theta + \cos^2 \theta)$? (3 Points)

Answer

Since $\sin^2 \theta + \cos^2 \theta = 1$ by trigonometric identities, $2(\sin^2 \theta + \cos^2 \theta)$ is simply 2.

4. If $\sin(\theta) = \frac{12}{13}$, and θ lies in the second quadrant, find the value of $\sec(\theta) + \tan(\theta)$. (8 Points)

Answer

As $\sin^2 \theta + \cos^2 \theta = 1$, $\cos \theta = \pm \sqrt{1 - \sin^2 \theta}$. Note that θ is mentioned to be in the second quadrant, so $\cos \theta$ is negative. Hence $\cos \theta = -\sqrt{1 - \sin^2 \theta}$.

Then

$$\sec \theta + \tan \theta = \frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta} = \frac{1 + \sin \theta}{\cos \theta} \quad (1)$$

$$= \frac{1 + \sin \theta}{-\sqrt{1 - \sin^2 \theta}} = \frac{1 + \frac{12}{13}}{-\sqrt{1 - (\frac{12}{13})^2}} = \frac{\frac{25}{13}}{-\sqrt{\frac{25}{169}}} = -5 \quad (2)$$